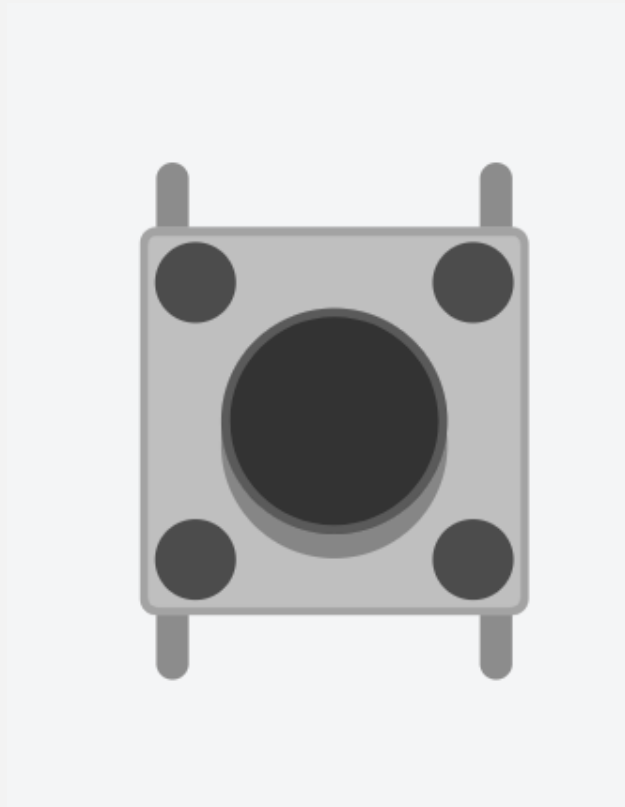
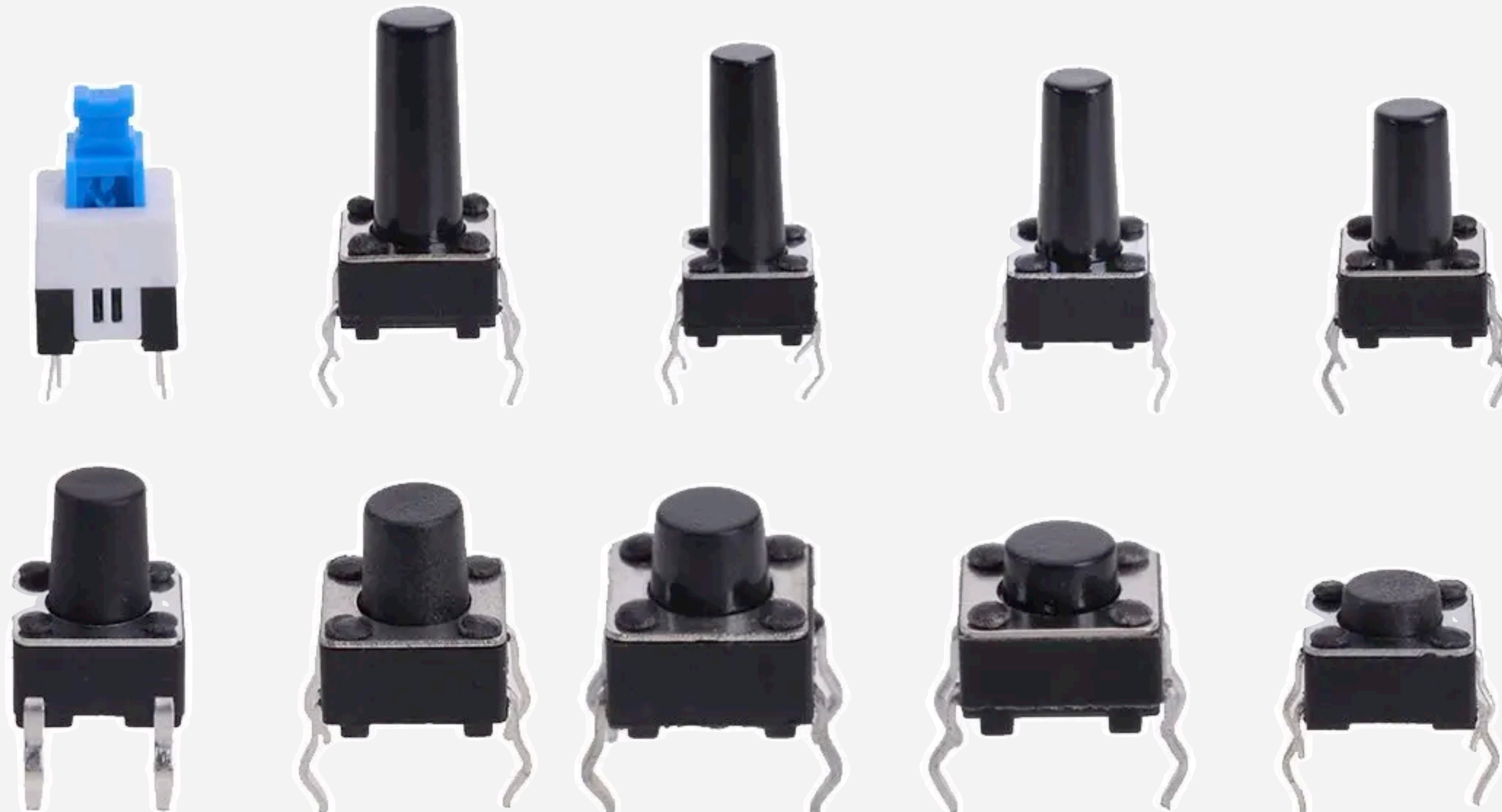


DIGITAL INPUT

Buttons and Switches



Buttons and Switches



Buttons and Switches



Buttons and Switches



Buttons and Switches

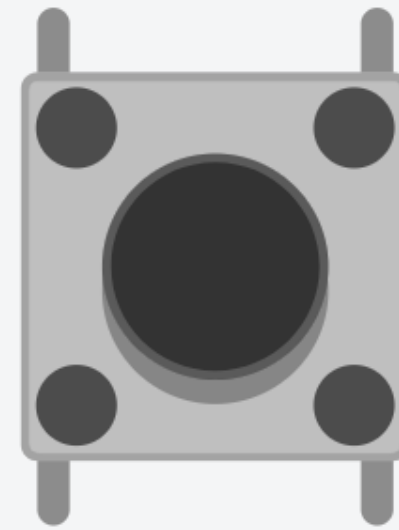
Buttons and Switches

A switch is any simple device that can complete or break an electrical connection.

Buttons and Switches

Anything can become a switch.

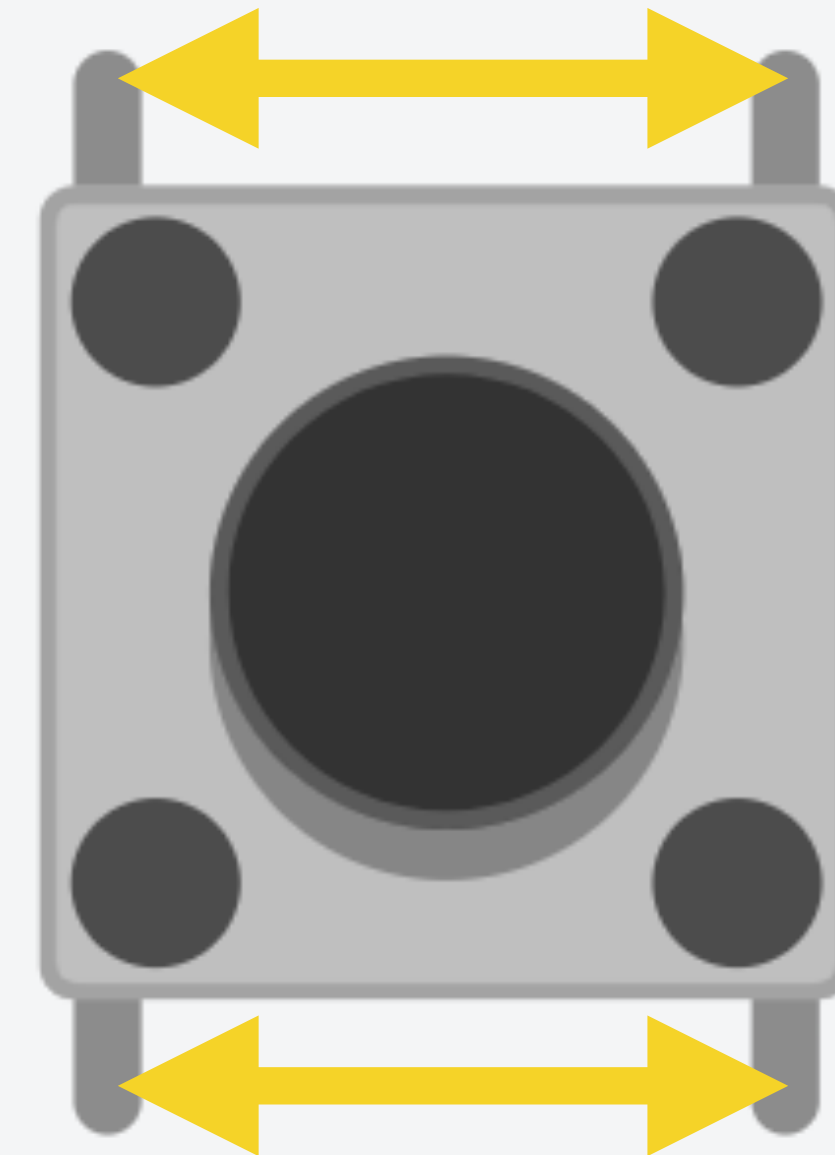
Buttons and Switches



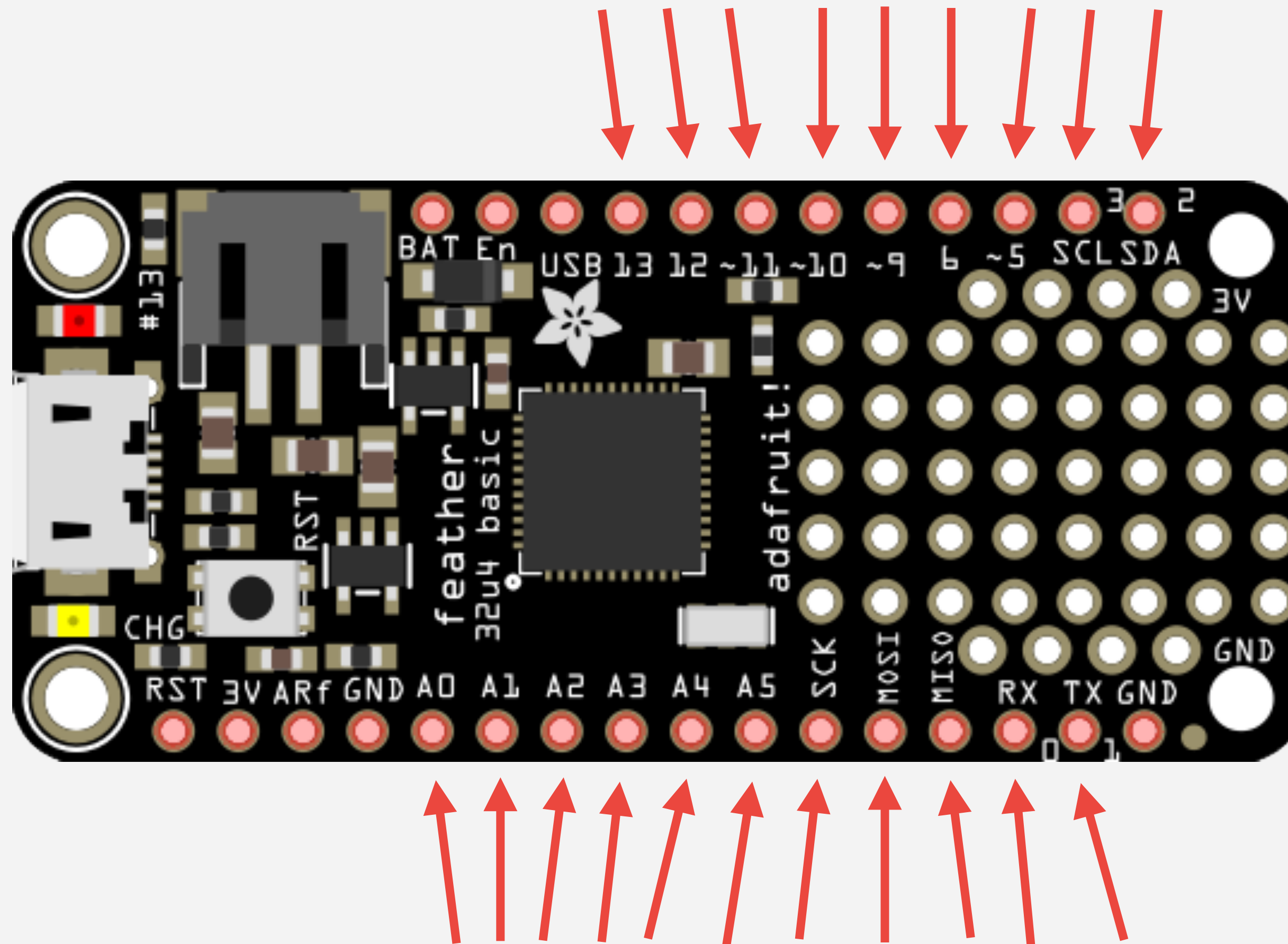
Buttons typically come in two types normally closed (NC) and normally open (NO). NO buttons complete the circuit, NC buttons break the circuit when pressed.

Buttons and Switches

When pressed, a NO button connects the legs from one side to those on the other side. Most of the buttons we will use are NO.



Using Buttons and Switches with the Arduino



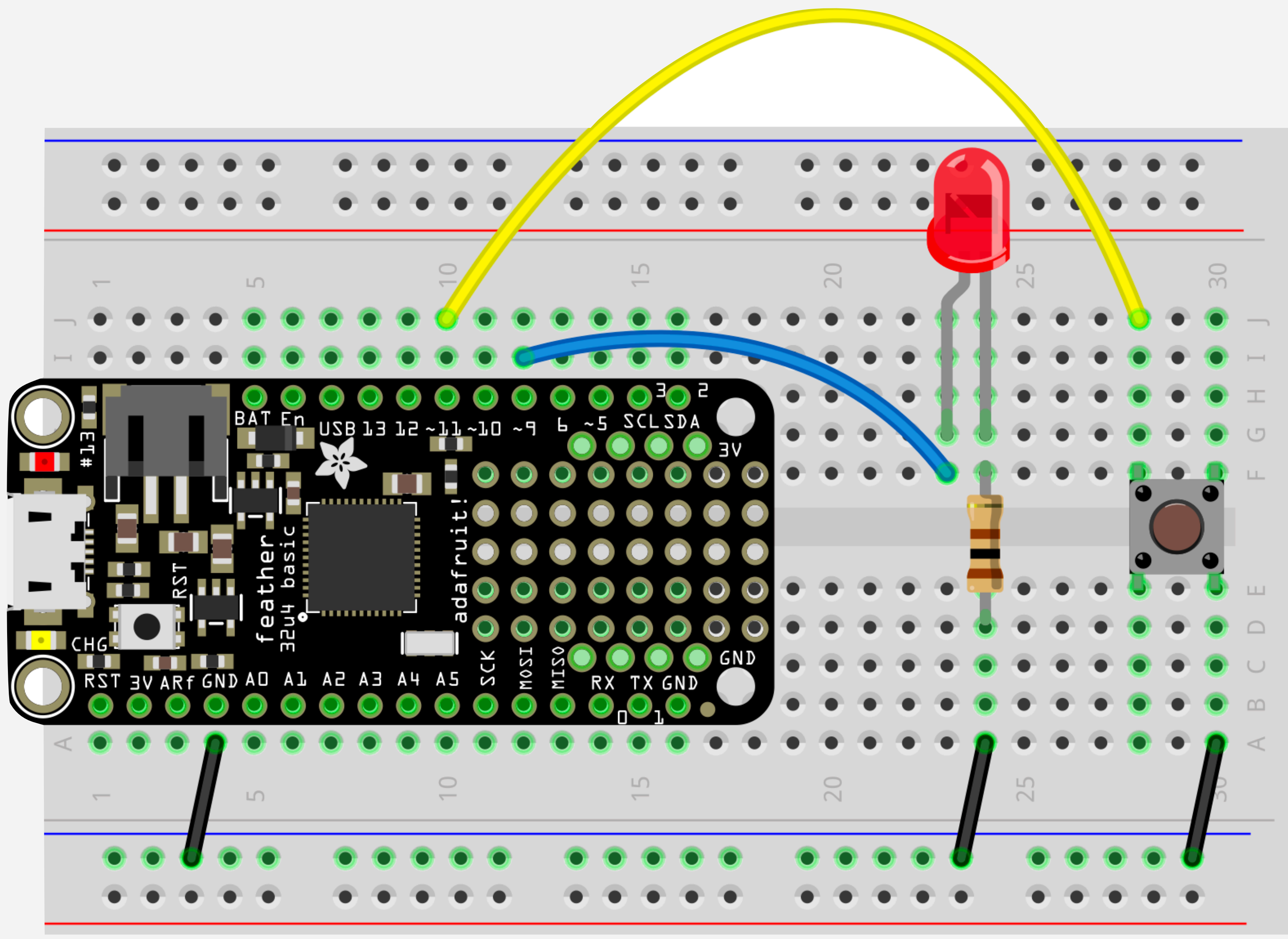
Digital Input and Output

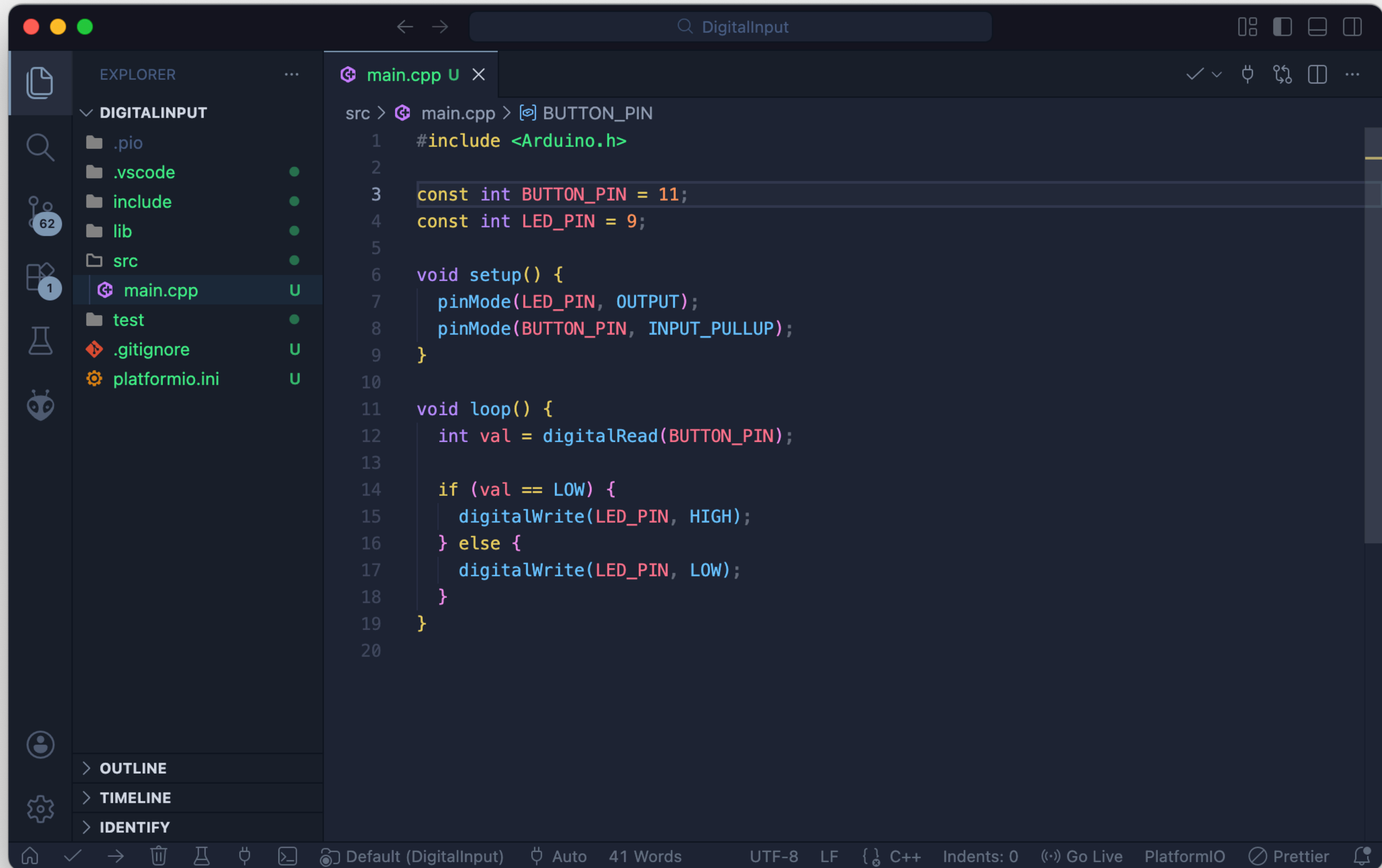
Arduino Input

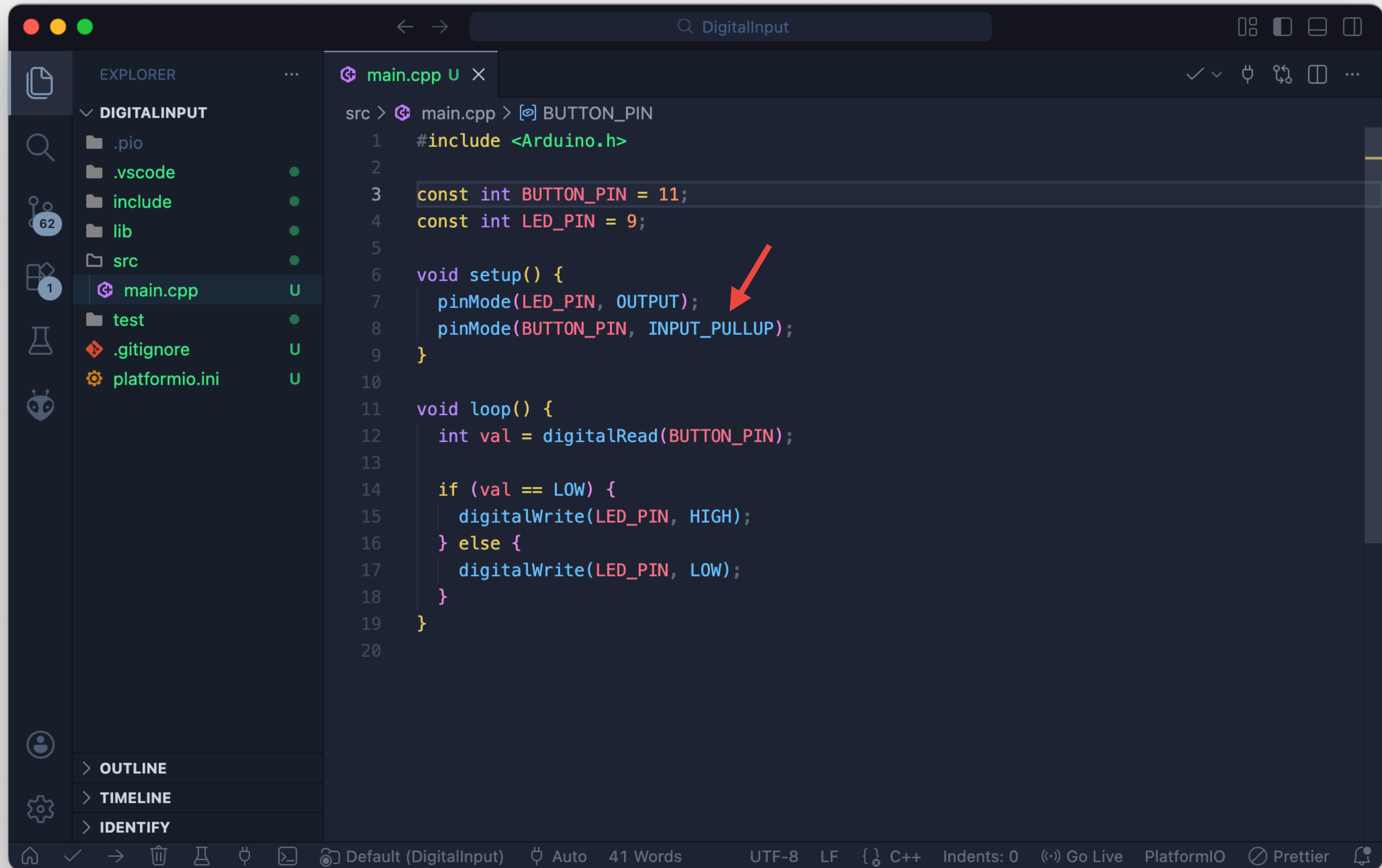
The inputs on the Arduino read voltage.

All inputs need to be thought of in terms of voltage differentials.

Digital inputs understand voltage levels above 2 volts as HIGH and less as LOW.







Try...

...having the LED make a pattern when the button is pressed.

...controlling two LEDs with one button.

...using two buttons that both must be pressed for the LED to turn on.

...using buttons from the bin in the FabLab.